Linear Algebra — Model Test 3: Answers and Methods

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: hard

Key methods used in this test

- **Proofs with dot products (1.2):** expand $\|x\|^2 = x \cdot x$ using the rules $(u+v)\cdot w = u\cdot w + v\cdot w$, $u \cdot v = v \cdot u$.
- **Rank-one matrices (2.4):** if $A = \alpha\beta^T$ (column times row) then $A^2 = (\beta^T\alpha)A$ and $A^n = (\beta^T\alpha)^{n-1}A$.
- **Elimination with a parameter (2.2):** never divide by an expression that might be 0; treat those values separately (zero pivot $\Rightarrow$ no solution or infinitely many).
- **Matrix equations (2.5):** collect the X terms: $(A-2I)X = A \Rightarrow X = (A-2I)^{-1}A$.
- $A = LU$ **(2.6):** the pivots are the diagonal of U; A has a full set of pivots iff no pivot is 0.
- **Proofs of invertibility (2.5):** exhibit B with $AB = BA = I$.

1 Vectors

(a)

Method. Expand both squared lengths as dot products.

$$\|u+v\|^2 = (u+v)\cdot(u+v) = \|u\|^2 + 2u\cdot v + \|v\|^2, \qquad \|u-v\|^2 = \|u\|^2 - 2u\cdot v + \|v\|^2.$$

Adding, the $\pm 2u\cdot v$ terms cancel: $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$. ■ (Geometrically, this is the parallelogram law: the squares of the two diagonals add up to the squares of the four sides.)

(b)

Method. $w = (x, y, z)$ must satisfy two dot-product equations; solve by elimination.

$x + y + z = 0$ and $x + 2y + 3z = 0$. Subtracting: $y + 2z = 0 \Rightarrow y = -2z$, then $x = -y - z = z$. So $w = z(1, -2, 1)$.

Answer. All multiples $w = t(1,-2,1)$, $t \in \mathbb{R}$ (a line through the origin).

(c) (i) $\cos\theta = \dfrac{(1,1,1)\cdot(1,0,0)}{\sqrt{3}\cdot 1} = \dfrac{1}{\sqrt{3}}$ ($\theta \approx 54.7°$). (ii) $\cos\theta = \dfrac{2}{\sqrt{3}\sqrt{2}} = \dfrac{\sqrt{6}}{3}$ ($\theta \approx 35.3°$).

Answer. (i) $1/\sqrt{3}$; (ii) $\sqrt{6}/3$.

2 Matrix powers and polynomials

(1)(a)

Method. Every row of A is a multiple of $(3,1,2)$: $A = \alpha\beta^T$ with $\alpha = (1,2,-1)^T$, $\beta^T = (3,1,2)$, and $\beta^T\alpha = 3 + 2 - 2 = 3$.

$A^2 = \alpha(\beta^T\alpha)\beta^T = 3A$ (direct multiplication confirms $A^2 = \begin{pmatrix} 9 & 3 & 6 \\ 18 & 6 & 12 \\ -9 & -3 & -6 \end{pmatrix}$). By induction $A^n = 3^{n-1}A$, so $A^{10} = 3^9 A = 19683A$.

Answer. $A^{10} = 19683A = \begin{pmatrix} 59049 & 19683 & 39366 \\ 118098 & 39366 & 78732 \\ -59049 & -19683 & -39366 \end{pmatrix}$

(b)

Method. I commutes with A; use the binomial theorem with $A^k = 3^{k-1}A$.

$(I+A)^n = I + \sum_{k=1}^{n} \binom{n}{k} 3^{k-1}A = I + \dfrac{4^n - 1}{3}A$. For $n = 5$: $\dfrac{1023}{3} = 341$.

Answer. $(I+A)^5 = I + 341A = \begin{pmatrix} 1024 & 341 & 682 \\ 2046 & 683 & 1364 \\ -1023 & -341 & -681 \end{pmatrix}$

(2)(a) $J^2 = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, $J^4 = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix} = -4I$, $J^8 = 16I$, so $J^{10} = 16J^2$.

Answer. $J^{10} = \begin{pmatrix} 0 & -32 \\ 32 & 0 \end{pmatrix}$

(b) $f(J) = J^4 - 2J^2 + 3I = -4I - 2J^2 + 3I = -I - 2J^2$.

Answer. $f(J) = \begin{pmatrix} -1 & 4 \\ -4 & -1 \end{pmatrix}$

3 Elimination with a parameter

Method. Exchange rows so the first pivot is 1, eliminate, and only divide by $\lambda - 1$ or $\lambda + 2$ after treating the cases where they vanish.

Swap $R_1 \leftrightarrow R_3$, then $R_2 - R_1$, $R_3 - \lambda R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & \lambda & \lambda^2 \\ 1 & \lambda & 1 & \lambda \\ \lambda & 1 & 1 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & \lambda & \lambda^2 \\ 0 & \lambda-1 & 1-\lambda & \lambda-\lambda^2 \\ 0 & 1-\lambda & 1-\lambda^2 & 1-\lambda^3 \end{array}\right).$$

Case $\lambda = 1$. Rows 2 and 3 become zero. Only $x_1 + x_2 + x_3 = 1$ remains, so there are infinitely many solutions: $x_1 = 1 - s - u,\ x_2 = s,\ x_3 = u$.
Case $\lambda \neq 1$. Divide Row 2 by $\lambda - 1$ and Row 3 by $1 - \lambda$, then $R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & \lambda & \lambda^2 \\ 0 & 1 & -1 & -\lambda \\ 0 & 1 & 1+\lambda & 1+\lambda+\lambda^2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & \lambda & \lambda^2 \\ 0 & 1 & -1 & -\lambda \\ 0 & 0 & \lambda+2 & (\lambda+1)^2 \end{array}\right).$$

$\lambda = -2$: the third pivot is 0 and the last equation reads $0 = 1$, so there is *no solution.*
$\lambda \neq 1, -2$: three pivots, unique solution. Back-substitute: $x_3 = \frac{(\lambda+1)^2}{\lambda+2}$, $x_2 = x_3 - \lambda = \frac{1}{\lambda+2}$, $x_1 = \lambda^2 - x_2 - \lambda x_3 = -\frac{\lambda+1}{\lambda+2}$. (Check $\lambda = 0$: $x = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ satisfies all three equations ✓.)

Answer. $\lambda \neq 1, -2$: unique solution $\left(-\frac{\lambda+1}{\lambda+2}, \frac{1}{\lambda+2}, \frac{(\lambda+1)^2}{\lambda+2}\right)$; $\lambda = -2$: no solution; $\lambda = 1$: infinitely many, $(1-s-u,\ s,\ u)$.

4 Inverses

(1)

Method. Row-reduce $(P \mid I_4)$.

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 3 & -2 & -1 & 3 & 0 & 1 & 0 & 0 \\ 3 & -1 & 1 & 2 & 0 & 0 & 1 & 0 \\ 1 & -1 & -2 & 3 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2-3R_1,\ R_3-3R_1,\ R_4-R_1} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -2 & -4 & 3 & -3 & 1 & 0 & 0 \\ 0 & -1 & -2 & 2 & -3 & 0 & 1 & 0 \\ 0 & -1 & -3 & 3 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3,\ -R_2,\ R_3+2R_2,\ R_4+R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & -2 & 3 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & 1 & -2 & 0 \\ 0 & 0 & -1 & 1 & 2 & 0 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{R_3\leftrightarrow R_4,\ -R_3,\ R_1-R_3,\ R_2-2R_3} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 3 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 & 7 & 0 & -3 & 2 \\ 0 & 0 & 1 & -1 & -2 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 3 & 1 & -2 & 0 \end{array}\right)$$

$$\xrightarrow{-R_4,\ R_1-R_4,\ R_3+R_4} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 6 & 1 & -3 & 1 \\ 0 & 1 & 0 & 0 & 7 & 0 & -3 & 2 \\ 0 & 0 & 1 & 0 & -5 & -1 & 3 & -1 \\ 0 & 0 & 0 & 1 & -3 & -1 & 2 & 0 \end{array}\right)$$

Answer. $P^{-1} = \begin{pmatrix} 6 & 1 & -3 & 1 \\ 7 & 0 & -3 & 2 \\ -5 & -1 & 3 & -1 \\ -3 & -1 & 2 & 0 \end{pmatrix}$

(2)

Method. $AX = A + 2X \Rightarrow (A-2I)X = A$, with $A - 2I = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$.

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3-R_1} \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3,\ R_1+R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow{R_1-R_3,\ R_2-R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

$$X = (A-2I)^{-1}A = \begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 3 & -1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 2 \\ -2 & -1 & 2 \\ 0 & 2 & 1 \end{pmatrix}.$$

Answer. $X = \begin{pmatrix} 1 & -2 & 2 \\ -2 & -1 & 2 \\ 0 & 2 & 1 \end{pmatrix}$

5 $A = LU$ with parameters

(a)

Method. Eliminate with multipliers; if $a \neq 0$ then $\ell_{21} = \ell_{31} = 1$.

$$\begin{pmatrix} a & a & a \\ a & b & b \\ a & b & c \end{pmatrix} \xrightarrow{R_2-R_1,\ R_3-R_1} \begin{pmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & b-a & c-a \end{pmatrix} \xrightarrow{R_3-R_2} \begin{pmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & 0 & c-b \end{pmatrix} = U,$$

with $\ell_{32} = 1$ (when $b \neq a$). So $L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$. The pivots are a, $b-a$, $c-b$.

Answer. L as above, $U = \begin{pmatrix} a & a & a \\ 0 & b-a & b-a \\ 0 & 0 & c-b \end{pmatrix}$; three pivots $\iff a \neq 0,\ a \neq b,\ b \neq c$.

(b) $U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$. $Lc = b$: $c_1 = 1$, $c_2 = 3-1 = 2$, $c_3 = 7-1-2 = 4$. $Ux = c$: $2x_3 = 4 \Rightarrow x_3 = 2$; $x_2 + x_3 = 2 \Rightarrow x_2 = 0$; $x_1 + x_2 + x_3 = 1 \Rightarrow x_1 = -1$. Check: $Ax = (-1+0+2,\ -1+0+4,\ -1+0+8) = (1, 3, 7)$ ✓

Answer. $x = (-1, 0, 2)^T$.

6 Proofs

(a) From $A^2 - A - 2I = 0$: $A(A-I) = A^2 - A = 2I$. So $A \cdot \frac{1}{2}(A-I) = I$ and $\frac{1}{2}(A-I) \cdot A = I$ (the factors commute). Hence

$$A^{-1} = \tfrac{1}{2}(A-I), \qquad (A-I)^{-1} = \tfrac{1}{2}A. \qquad \blacksquare$$

(b) Let $S = I + N + \cdots + N^{k-1}$. Then $(I-N)S = (I+N+\cdots+N^{k-1}) - (N+\cdots+N^k) = I - N^k = I$, and likewise $S(I-N) = I$. So $(I-N)^{-1} = I + N + \cdots + N^{k-1}$. $\blacksquare$